

Chapter 1

Literature Review

In this chapter, I aim to explore the existing literature that delves into the fundamentals of the various elements involved within this project, starting with [Radio Detection and Ranging \(RADAR\)](#), and more specifically, [Active Radar \(AR\)](#). Then I delve into [Orthogonal Frequency Division Multiplexing \(OFDM\)](#), followed by [Software Defined Radio \(SDR\)](#) and its working and uses within the world of RADAR. Then, I look into OFDM based AR, and finally, I look into OFDM based classification RADAR. My aim for this review of the literature is to add context to the broader paper, and hopefully give you, the reader, a more in depth understanding of the core concepts one has to understand to fully grasp the overall workings of the project. Especially as we move towards the latter parts of the project, where classification is involved, it is important to grasp the fundamentals.

I will separate the literature review into distinct sections, where I will explore the history behind each concept and the theory involved within. I will also explore and analyze the advantages and disadvantages of each topic and then finally, provide a critique of the literature towards the end.

1.1 Classical RADAR

1.1.1 Brief History

The fundamental idea behind RADAR systems and the technology itself were first theorized during the classical electromagnetic radiation experiments of the 1880s performed by Heinrich Hertz [1]. These experiments set out to prove the previous theoretical work achieved by James Clerk Maxwell. Maxwell came up with the formulae that described electromagnetic fields, and he hypothesized that this formula would apply to both light and radio waves since they were both examples of electromagnetic radiation just at vastly different frequencies. This has obviously since been proved, with visible light roughly falling into the 400THz to 700THz frequency range. The work done by Maxwell led to the discovery that electromagnetic waves can be reflected by metallic objects and refracted by a dielectric medium, just like light can [1]. This work did not go unnoticed and a patent for an ‘an obstacle detector and ship navigation device’ was issued in 1904 to Christian Hülsmeyer, a German engineer [1]. Work on RADAR then slowed down due to a lack of interest from governmental authorities until the early 1930s, when the development of long-range bombers prompted the major powers of the world to develop technologies which could detect these hostile aircraft, which would be useful for the coming second world war.

1.1.2 RADAR Workings

At its core, a RADAR is defined as an instrument that can detect surrounding objects using radio waves. To achieve this, there are multiple different setups of RADAR systems, which will be discussed later. These RADAR setups however all rely on the following fundamental laws of physics which are listed below [2]:

1. **Reflection of electromagnetic waves:** This rule describes how a radio wave will be reflected back towards a transmitter of said radio wave if the object is placed directly in front of the transmitter. This reflection indicates that the object is in front of the transmitter.
2. **Electromagnetic energy travels through air at a constant speed:** This rule describes how the electromagnetic waves transmitted by an antenna propagate through the transmission medium, usually air for our earth's use cases, at roughly the speed of light, c . This velocity c equates to $299,792,458 \text{ m.s}^{-1}$
3. **Electromagnetic energy normally travels through space in a straight line:** This describes how the energy transmitted by an antenna in a radar system propagates through space in a straight line. This is important because it means that the energy of the RADAR system can be directed in a specific direction by the antenna system, which means that the reflecting object's direction (azimuth and elevation) can be determined.

In a classic pulsed RADAR, essentially the most simple kind of RADAR that is used today, by using the above-mentioned physical properties, the distance, azimuth, and elevation of the object that is being observed by the radar system can be determined. The best example of this sort of RADAR system is an airport RADAR. An [Airport Surveillance RADAR \(ASR\)](#) is used at airports to determine the presence and position of aircraft in the vicinity of an airport. Typically, ASR systems are capable of reliably detecting and tracking aircraft at altitudes below 25,000 feet (7,620 meters) and within 40 to 60 nautical miles (75 to 110 km) of the airport [3].

Classical RADAR systems can be separated into a few different categories [4]. Broadly, you get primary and secondary RADARs that are separated into imaging and non-imaging RADARs. An imaging RADAR is one which desires to draw an image, such as a weather RADAR, and a non-imaging RADAR is one which does not draw an image but rather infers information about a target based on processing techniques. Primary RADARs rely on echos from a target that come from it's own transmitted pulse, whereas a secondary radar relies on a RADAR response signal from a target. Sometimes, the ASR systems mentioned previously have secondary radar systems built-in which allow the aircraft being detected to broadcast vital information to aid with air traffic control. This is usually only possible for 'friendly' targets. The various categorizations of RADAR types can be seen in figure 1.1 below:

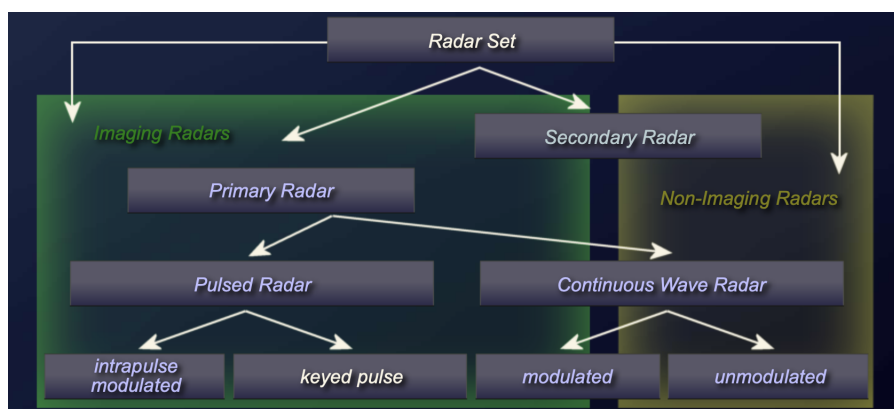


Figure 1.1: Classification of Radar Systems [4]

Note to self, I don't think I like this image

1.1.3 Advantages and Disadvantages of RADAR

Since RADAR utilizes the electromagnetic spectrum, there are distinct advantages concerning the use of RADAR systems [2]. The main advantage is that, unlike humans, RADAR systems can operate day and night, 24/7, over far distances, in good or bad weather. Compared to human eyes, this is obviously a lot better. Some other advantages of classic RADAR is that it can generally 'see' through insulators, and it can also distinguish the difference between fixed and moving targets [5]. Some disadvantages however, are that classical RADAR systems generally struggle to resolve targets that are very close to the transmitter, and they cannot penetrate water very well [5]. Classical RADAR systems also fail to distinguish between shapes of object that they are looking at.

1.2 OFDM

1.2.1 Brief History

The general working principles, and the mathematics required for OFDM were originally described by Robert Chang in 1966 while he was working at Bell Labs [6]. While the idea of OFDM was smart, it didn't become popular until sufficiently cheap microprocessors entered the market [6]. OFDM was never implemented in commercial communication channels until the appearance of mass market applications coincided with the availability of efficient software and electronic technologies [7].

1.2.1.1 Previous Transmission Techniques

Before OFDM there was [Frequency Division Multiplexing \(FDM\)](#). FDM system innovation was driven out of necessity due to telegraph companies hoping to increase the amount of users that could utilize their transmission lines to increase profits from their systems [7]. Transmission companies were in search of a method to have multiple messages being sent on the same transmission line at simultaneously. Other methods had been used such as [Time Division Multiplexing \(TDM\)](#), and [Time Division Multiple Access \(TDMA\)](#). These methods were however limited by burst speed which was itself limited by inter-symbol interference. Around the 1920s, FDM became the main carrier system for telephone companies. The

first major company to deploy the system was [American Telephone and Telegraph Company \(AT&T\)](#). This initial deployment of the system used 8kHz subchannels with 4kHz allocated to each channel. This form of transmission system later developed into systems that had more subcarrier channels with less subcarrier spacing. There were however some drawbacks to FDM, which are highlighted below:

1. **Waste of frequency:** Due to the large subcarrier spacing, FDM wasted a lot of precious frequency spectrum to these so-called ‘guard’ bands on either side of the carrier. As the frequency spectrum became more crowded throughout the years, these guard bands became a liability due to demand for spectrum being too high, and the guard bands were not efficiently using the spectrum.
2. **Using different modulators for the various sub-channels:** In the old systems, each sub-channel had a separate modulator in the up-mixing process. This multiplication process became tedious once there were multiple sub-carriers.

To compensate for this inefficient use of the frequency spectrum, OFDM was introduced. What made OFDM different to FDM was that each subcarrier overlapped some of the other carriers in signal. This was able to be done because each carrier was spaced in such a way that it was orthogonal to the other carriers in the signal [7].

1.2.1.2 OFDM

As mentioned before, [OFDM](#) is a transmission method that uses the spectrum in a more efficient manner to transmit as much information as possible over a single transmission line. The transmission of information is commonly done using various modulation schemes such as QAM encoding that breaks information into bit chunks that are sent in parallel over multiple channels [8]. The OFDM transmission scheme builds upon FDM where multiple channels are created over a specific bandwidth, and bits of information are sent on each channel in parallel. The main difference between FDM and OFDM is therefore the orthogonality of the OFDM signal. I have created a crude example below in figure 1.2. In fig 1.2a, a crude mockup of an OFDM signal can be seen in the frequency domain. As can be seen, all the peaks of the OFDM waveform have zero interference with any other part of the signal. This is due to the various carriers being orthogonal to each other. Conversely, with the FDM waveform, the peaks have significant interference from the other channels since the channels are not spaced orthogonally in regard to each other.

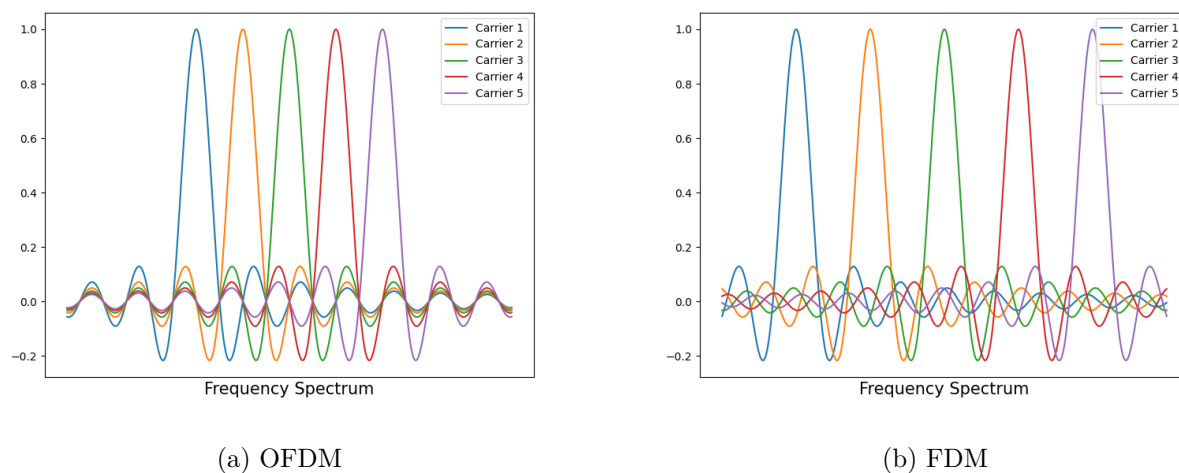


Figure 1.2: OFDM and FDM Waveforms

I want to touch on the idea of orthogonality. In practice, all it means is that in a signal with multiple sinusoidal carriers, each carrier has a frequency that is a whole number multiple of some fundamental frequency [6]. The time domain voltage signal of an OFDM signal can be described by equation 1.1 below:

$$v(t) = \sum_{k=0}^{N-1} X_k e^{j2\pi kt/T}, \quad 0 \leq t < T \quad (1.1)$$

N = Number of subcarriers

where X_k = Data symbols in the frequency domain

T = OFDM symbol time (s)

The subcarrier spacing of $\frac{1}{T}$ ensures carrier orthogonality over the OFDM frame, or symbol. In practice, real world implementations of OFDM signals will have vastly more complex components added on concerning error correction and data modulation techniques in general. The base signal however will still revolve around the idea of orthogonal carriers.

1.2.2 Advantages and Disadvantages of OFDM

As can be seen in the above sections, OFDM is a very useful and versatile transmission technique that is used in a variety of real world applications. While OFDM is a widely used and very good transmission technique, it does come with some pros and cons [9]. The biggest pro for OFDM, particularly with its use within the telecommunications industry, is the spectral efficiency that can be achieved. Real world OFDM systems are also resistant to multipath fading due to the cyclic prefix that these systems implement. Modern OFDM systems can allow for easy equalization with embedded pilot symbols. Equalization is much harder to achieve in a single carrier system. Some cons of OFDM systems however include the following. Firstly, an OFDM transmission can have a high [Peak to Average Power Ratio](#)

(PAPR). This is due to an OFDM signal's noise-like amplitude with a large dynamic range [9]. Another problem is that OFDM systems are susceptible to carrier frequency offset due to local oscillators not being able to perfectly operate at the same frequency on the transmitter and receiver.

1.3 OFDM based RADAR

1.3.1 Brief History of OFDM based RADAR

1.3.2 Implementation of OFDM based RADAR

There is an ever evolving area of RADAR research that involves the combination of RADAR systems and telecommunication systems. This is colloquially known as [RADAR Communications \(Radcom\)](#). Radcom aims to combine radar and telecommunications because this could potentially lead to system that are more spectrally efficient, and lightweight. In essence, one could imagine a RADAR system that can communicate with other base stations, as well as performing usual RADAR tasks, all from a single system. An attempt at creating a system like this was attempted by Dr Stephen Paine and Grant Norrie in [10]. They set out to create a RADAR system that utilizes the previously mentioned OFDM signal structure in a radcom setup. They wanted to base the system around an OFDM signal because of the ever-increasing popularity of OFDM signals in communication networks. They mention that communication signals are often used in passive RADARs, by exploiting [Illuminators of Opportunity \(IOO\)](#). An IOO is a transmitter that is not intended for RADAR purposes, but rather other applications, with the most popular being telecommunications purposes. They pose the question, if a passive RADAR system can use existing communication signals to detect objects, why can an active system not do the same?

They explored the use of the OFDM based [Digital Audio Broadcasting \(DAB\)](#) modulation scheme as the backbone of the system. Part of the reason for this, is the fact that DAB is a very popular and well-defined. They designed and documented an entire toolchain that consisted of the following parts.

1. Signal Generation Chain: This entailed the building of the signal that was to be transmitted by their receiver.
2. Hardware Testbed: This was the physical hardware that was used to implement the theoretical in the real world. This consisted of dual Ettus USRP N210 SDRs.
3. Radar Processing Chain: Everything to do with processing of the signal on the reception side of things to verify the operation of the system as a RADAR. The main objective of this toolchain was to generate a range-doppler graph to visualize the output of the overall RADAR system.
4. Communications Processing Chain: This toolchain was in place to verify and perform the necessary operations to facilitate a working communication link between the transmitter and the receiver.

The SDR platform had specific configuration characteristics that were used for the tests performed by [10]. These settings are summarized in table 1.1 below:

Parameter	Value
Tx Gain	0 dB
Rx Gain	0 dB
Tx Bandwidth	1.1 MHz
Rx Bandwidth	1.1 MHz
Sampling Rate	5 MHz
Carrier Frequency	2.45 GHz

Table 1.1: SDR Toolchain Parameters

Multiple tests were performed to test the capability of the system. These tests were a communication test, a static range test, and a range-doppler test, which involved a moving target. The communication test was fairly basic, and it aimed to eliminate antenna/system feed through, and then measured the [Bit Error Rate \(BER\)](#) upon transmission from the transmit antenna to the reception antenna from a distance of 5 meters away. The BER was confirmed to be 0% and then the test was deemed to be a success.

Then a static range test was performed with an object 100m away from the RADAR. After pulse cancellation with adjusted dynamic range, and clutter map removal, a range-doppler map was generated as shown below in figure 1.3:

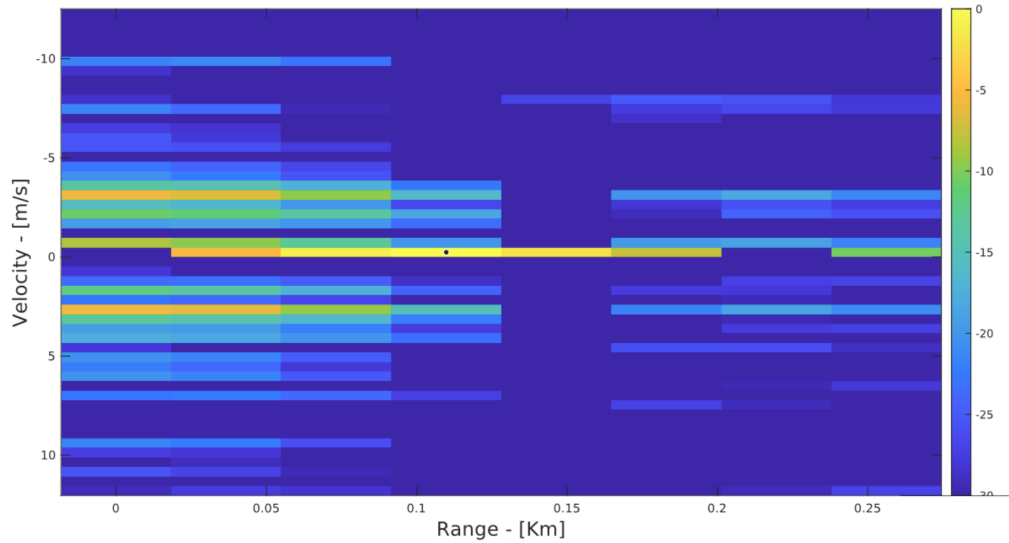


Figure 1.3: Range-doppler map of object at 100m

This figure showed that the object was detectable at 100m, specifically, the object could be seen in the 109m range bin. Lastly, a moving target test was performed, which was able to successfully detect a moving car. The ground truth of the moving object was as follows, it was 30m away from the RADAR and was moving at a velocity of 7.2 m.s^{-1} and 7.0 m.s^{-1} as measured by a [Global Positioning System \(GPS\)](#) and the odometer respectively. The measured range and velocity was determined to be 36m and -7.4 m.s^{-1} . The negative velocity is due to the object moving away from the RADAR system.

In [11], Gabriel Lellouch

1.3.3 Advantages and Disadvantages of OFDM based RADAR

1.4 Critique of Literature

Notes to self

....In Stephen Paine and Norrie paper, the communication test is quite trivial...

Bibliography

- [1] M. Skolnik. Radar - detection, military, technology | britannica. [Online]. Available: <https://www.britannica.com/technology/radar/History-of-radar>
- [2] Christian Wolff. Physical fundamentals of the radar principle. [Online]. Available: <https://www.radartutorial.eu/01.basics/Physical%20fundamentals%20of%20the%20radar%20principle.en.html>
- [3] everythingRF. What is an airport surveillance radar (ASR) system? [Online]. Available: <https://www.everythingrf.com/community/what-is-an-airport-surveillance-radar-asr-system>
- [4] Christian Wolff. Classification of radar systems (1). [Online]. Available: <https://www.radartutorial.eu/02.basics/Classification%20of%20Radar%20systems%20%281%29.en.html>
- [5] A. Mishra. Radar | types of radar, advantages and disadvantages. [Online]. Available: <https://www.engineeringa2z.com/radar-types-of-radar-advantage-disadvantage/>
- [6] C. S. Lessons. (2024) Wireless communication nine: Ofdm. YouTube video. [Online]. Available: <https://www.youtube.com/watch?v=OLsbONSQFUI>
- [7] S. B. Weinstein, “The history of orthogonal frequency-division multiplexing.” [Online]. Available: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=5307460>
- [8] G. Wright. What is orthogonal frequency-division multiplexing (OFDM)? [Online]. Available: <https://www.techtarget.com/searchnetworking/definition/orthogonal-frequency-division-multiplexing>
- [9] R. W. World. Ofdm: Advantages and disadvantages of orthogonal frequency division multiplexing. Article. [Online]. Available: <https://www.rfwireless-world.com/terminology/ofdm-advantages-disadvantages>
- [10] G. Norrie and S. Paine, “Design and demonstration of an ofdm based radcom system,” in *2023 IEEE Radar Conference (RadarConf23)*, 2023, pp. 1–6.
- [11] G. Lellouch, “WAVEFORM DESIGN AND PROCESSING TECHNIQUES IN OFDM RADAR.”

Glossary

AR Active Radar

ASR Airport Surveillance RADAR

AT&T American Telephone and Telegraph Company

BER Bit Error Rate

DAB Digital Audio Broadcasting

FDM Frequency Division Multiplexing

GPS Global Positioning System

IOO Illuminators of Opportunity

OFDM Orthogonal Frequency Division Multiplexing

PAPR Peak to Average Power Ratio

RADAR Radio Detection and Ranging

Radcom RADAR Communications

SDR Software Defined Radio

TDM Time Division Multiplexing

TDMA Time Division Multiple Access